

Everything below is a *read* of one binary merge tree per jet. Clustering records, per step: **hist_p1,p2** (the two parent ids merged; p2=-1 beam merge = jet finished, -2 pad), **hist_child** (new id), **hist_d** (the distance below). Initial particles are ids 0..*n*-1; each merge makes a new id. E-scheme $\Rightarrow$ a pseudojet's 4-momentum is the *sum* of its constituents (why **pseudojet.p4** is just addition).

0. The clustering (foundation — anti-kt / kt / C-A)

Sequential recombination: repeatedly merge the pair with the smallest distance, or merge *i* with the beam.

$$d_{ij} = \min(k_{t,i}^{2p}, k_{t,j}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \quad d_{iB} = k_{t,i}^{2p}, \quad \Delta R_{ij}^2 = \Delta y_{ij}^2 + \Delta \phi_{ij}^2.$$

The exponent *p* picks the algorithm: **kt** *p*=+1 (soft first), **Cambridge/Aachen** *p*=0 (purely angular, $d_{ij} = \Delta R^2/R^2$), **anti-kt** *p*=-1 (hard first, round jets). *y* = rapidity, *k_t* = transverse momentum. **hist_d** stores exactly this d_{ij} .
[*anti-kt 0802.1189 p2-3*]

1. Exclusive jets — exclusive_jets_from_history (a kt concept)

Inclusive clustering runs to the end (every particle in some jet). **Exclusive** = stop early and read off the clusters that remain. Two stopping rules:

- *d_{cut}*: undo every pair-merge with $d_{ij} \geq d_{cut}$ (keep $d_{ij} < d_{cut}$). **keep_pair** = **hist_d** < **d_{cut}**
- *n_{jets}*: keep only the first $k = n_{init} - n_{jets}$ merges of the sequence, leaving exactly *n_{jets}* clusters. (For kt, *d* is monotonic $\Rightarrow$ “first *k* steps” = “smallest-*d* *k* merges” = FastJet’s undo-the-widest-*k*.)

Then each particle is assigned to the highest pseudojet it reaches through *kept* merges only (**_resolve_roots**). Anchor: *n_{jets}* = *n_{inclusive}* reproduces the inclusive partition. [FastJet manual p14, p20]

2. Grooming — groom_from_history (soft-drop / mass-drop)

Goal: strip soft wide-angle radiation, keep the hard 2-prong core (cleaner jet mass). **Walk each jet from its root down the harder branch**, undoing the widest split first. At each node with parents *i, j* (let $z = \frac{\min(p_{t,i}, p_{t,j})}{p_{t,i} + p_{t,j}}$, $\Delta R = \Delta R_{ij}$), test the **soft-drop condition**

$$z > z_{cut} \left(\frac{\Delta R}{R} \right)^\beta.$$

- **passes** $\rightarrow$ this node is the groomed jet; stop.
- **fails** $\rightarrow$ drop the softer parent, recurse into the harder one.
- reach a single particle without passing $\rightarrow$ *untagged*.

Special cases: $\beta=0$ is the **modified Mass-Drop Tagger (mMDT)** (condition becomes just $z > z_{cut}$). Passing also $\mu=$ adds the **original mass-drop** requirement $\max(m_i, m_j) < \mu m_{node}$ (MDT default $\mu=0.67$, $y_{cut}=0.09$). The walk depth is the $\mathcal{O}(\log_2 n)$ tree descent. [BDRS 0802.2470; mMDT 1307.0007 p24-26; SoftDrop 1402.2657 p3,p6]

3. Lund coordinates — lund_coordinates_from_history

De-cluster each jet (widest split first) and, at every split, emit the kinematics of that emission — the inputs to the **primary Lund plane** (a standard substructure / ML representation). Per split, from the two parents *i, j*:

$$z = \frac{\min(p_{t,i}, p_{t,j})}{p_{t,i} + p_{t,j}}, \quad \Delta R = \Delta R_{ij}, \quad k_t = \min(p_{t,i}, p_{t,j}) \Delta R,$$

and the code stores the 6-tuple $(z, \Delta R, k_t, \ln \frac{1}{\Delta R}, \ln k_t, d)$; the Lund plane is the $(\ln \frac{1}{\Delta R}, \ln k_t)$ scatter of a jet’s splits. The last channel *d* equals **splitting_scales** exactly (the built-in cross-check). Bounds on real splits: $z \in (0, \frac{1}{2}]$, $\Delta R \leq R$.
[Lund plane 1807.04758 p4-6]

Mental model: clustering builds a binary tree per jet. **F1** cuts the tree at a chosen depth (exclusive jets); **F2** walks it top-down peeling soft branches (grooming); **F3** logs the physics of every split (Lund). The pages that follow are the primary-source definitions of each formula above.

A priori it is not clear whether it is better to have regular (‘soft-resilient’) or less regular (soft-adaptable) jets. In particular, regularity implies a certain rigidity in the jet algorithm’s ability to adapt a jet to the successive branching nature of QCD radiation. On the other hand knowledge of the typical shape of jets is often quoted as facilitating experimental calibration of jets, and soft-resilience can simplify certain theoretical calculations, as well as eliminate some parts of the momentum-resolution loss caused by underlying-event and pileup contamination.

Examples of jet algorithms with a soft-resilient boundary are the plain “iterative cone” algorithm, as used for example in the CMS collaboration [6], and fixed-cone algorithms such as Pythia’s [7] CellJet. The CMS iterative cone takes the hardest object (particle, calorimeter tower) in the event, uses it to seed an iterative process of looking for a stable cone, which is then called a jet. It then removes all the particles contained in that jet from the event and repeats the procedure with the hardest available remaining seed, again and again until no seeds remain. The fixed-cone algorithms are similar, but simply define a jet as the cone around the hardest seed, skipping the iterative search for a stable cone. Though simple experimentally, both kinds of algorithm have the crucial drawback that if applied at particle level they are collinear unsafe, since the hardest particle is easily changed by a quasi-collinear splitting, leading to divergences in higher-order perturbative calculations.¹

In this paper it is not our intention to advocate one or other type of algorithm in the debate concerning soft-resilient versus soft-adaptable algorithms. Rather, we feel that this debate can be more fruitfully served by proposing a simple, IRC safe, soft-resilient jet algorithm, one that leads to jets whose shape is not influenced by soft radiation. To do so, we take a quite non-obvious route, because instead of making use of the concept of a stable cone, we start by generalising the existing sequential recombination algorithms, k_t [1] and Cambridge/Aachen [2].

As usual, one introduces distances d_{ij} between entities (particles, pseudojets) i and j and d_{iB} between entity i and the beam (B). The (inclusive) clustering proceeds by identifying the smallest of the distances and if it is a d_{ij} recombining entities i and j , while if it is d_{iB} calling i a jet and removing it from the list of entities. The distances are recalculated and the procedure repeated until no entities are left.

The extension relative to the k_t and Cambridge/Aachen algorithms lies in our definition of the distance measures:

$$d_{ij} = \min(k_{ti}^{2p}, k_{tj}^{2p}) \frac{\Delta_{ij}^2}{R^2}, \quad (1a)$$

$$d_{iB} = k_{ti}^{2p}, \quad (1b)$$

where $\Delta_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$ and k_{ti} , y_i and ϕ_i are respectively the transverse momentum, rapidity and azimuth of particle i . In addition to the usual radius parameter R , we have added a parameter p to govern the relative power of the energy versus geometrical (Δ_{ij}) scales.

For $p = 1$ one recovers the inclusive k_t algorithm. It can be shown in general that for $p > 0$ the behaviour of the jet algorithm with respect to soft radiation is rather similar to that observed for the k_t algorithm, because what matters is the ordering between particles and for finite Δ this is maintained for all positive values of p . The case of $p = 0$ is special and it corresponds to the inclusive Cambridge/Aachen algorithm.

¹This is discussed in the appendix in detail for the iterative cone, and there we also introduce the terminology iterative cone with split–merge steps (IC-SM) and iterative cone with progressive removal (IC-PR), so as to distinguish the two broad classes of iterative cone algorithms.

Negative values of p might at first sight seem pathological. We shall see that they are not.² The behaviour with respect to soft radiation will be similar for all $p < 0$, so here we will concentrate on $p = -1$, and refer to it as the “anti- k_t ” jet-clustering algorithm.

2 Characteristics and properties

2.1 General behaviour

The functionality of the anti- k_t algorithm can be understood by considering an event with a few well-separated hard particles with transverse momenta $k_{t1}, k_{t2}, \dots$ and many soft particles. The $d_{1i} = \min(1/k_{t1}^2, 1/k_{ti}^2)\Delta_{1i}^2/R^2$ between a hard particle 1 and a soft particle i is exclusively determined by the transverse momentum of the hard particle and the Δ_{1i} separation. The d_{ij} between similarly separated soft particles will instead be much larger. Therefore soft particles will tend to cluster with hard ones long before they cluster among themselves. If a hard particle has no hard neighbours within a distance $2R$, then it will simply accumulate all the soft particles within a circle of radius R , resulting in a perfectly conical jet.

If another hard particle 2 is present such that $R < \Delta_{12} < 2R$ then there will be two hard jets. It is not possible for both to be perfectly conical. If $k_{t1} \gg k_{t2}$ then jet 1 will be conical and jet 2 will be partly conical, since it will miss the part overlapping with jet 1. Instead if $k_{t1} = k_{t2}$ neither jet will be conical and the overlapping part will simply be divided by a straight line equally between the two. For a general situation, $k_{t1} \sim k_{t2}$, both cones will be clipped, with the boundary b between them defined by $\Delta R_{1b}/k_{t1} = \Delta_{2b}/k_{t2}$.

Similarly one can work out what happens with $\Delta_{12} < R$. Here particles 1 and 2 will cluster to form a single jet. If $k_{t1} \gg k_{t2}$ then it will be a conical jet centred on k_1 . For $k_{t1} \sim k_{t2}$ the shape will instead be more complex, being the union of cones (radius $< R$) around each hard particle plus a cone (of radius R) centred on the final jet.

The key feature above is that the soft particles do not modify the shape of the jet, while hard particles do. I.e. the jet boundary in this algorithm is resilient with respect to soft radiation, but flexible with respect to hard radiation.³

The behaviours of different jet algorithms are illustrated in fig. 1. We have taken a parton-level event together with $\sim 10^4$ random soft ‘ghost’ particles (as in [4]) and then clustered them with 4 different jet algorithms. For each of the partonic jets, we have shown the region within which the random ghosts are clustered into that jet. For the k_t and Cambridge/Aachen algorithms, that region depends somewhat on the specific set of ghosts and the jagged borders of the jets are a consequence of the randomness of the ghosts — the jet algorithm is adaptive in its response to soft particles, and that adaptiveness applies also to the ghosts which take part in the clustering. For SISCone one sees that single-particle jets are regular (though with a radius $R/2$ — as pointed out in [4]), while composite jets have more varied shapes. Finally with the anti- k_t algorithm, the hard jets are all circular with a radius R , and only the softer jets have more complex shapes. The pair of jets near $\phi = 5$ and $y = 2$ provides an interesting example in this respect. The left-hand one is much softer than the right-hand one. SISCone (and Cam/Aachen) place the boundary between

²Note that, for $p < 0$, $\min(k_{ti}^{2p}, k_{tj}^{2p})$ differs from another possible extension, $[\min(k_{ti}^2, k_{tj}^2)]^p$, which *can* lead to strange behaviours.

³For comparison, IC-PR algorithms behave as follows: with $R < \Delta_{12} < 2R$, the harder of the two jets will be fully conical, while the softer will be clipped regardless of whether p_{t1} and p_{t2} are similar or disparate scales; with $\Delta_{12} < R$ the jet will be just a circle centred on the final jet.

3.3.2 Accessing exclusive jets

There are two ways of accessing exclusive jets, one where one specifies d_{cut} , the other where one specifies that the clustering is taken to be stopped once it reaches the specified number of jets.

```
/// return a vector of all jets (in the sense of the exclusive algorithm) that would
/// be obtained when running the algorithm with the given dcut.
vector<PseudoJet> exclusive_jets (const double & dcut) const;

/// return a vector of all jets when the event is clustered (in the exclusive sense)
/// to exactly njets. Throws an error if the event has fewer than njets particles.
vector<PseudoJet> exclusive_jets (const int & njets) const;

/// return a vector of all jets when the event is clustered (in the exclusive sense)
/// to exactly njets. If the event has fewer than njets particles, it returns all
/// available particles.
vector<PseudoJet> exclusive_jets_up_to (const int & njets) const;
```

The user may also wish just to obtain information about the number of jets in the exclusive algorithm:

```
/// return the number of jets (in the sense of the exclusive algorithm) that would
/// be obtained when running the algorithm with the given dcut.
int n_exclusive_jets (const double & dcut) const;
```

Another common query is to establish the d_{min} value associated with merging from $n + 1$ to n jets. Two member functions are available for determining this:

```
/// return the dmin corresponding to the recombination that went from n+1 to n jets
/// (sometimes known as  $d_{n,n+1}$ ).
double exclusive_dmerge (const int & n) const;

/// return the maximum of the dmin encountered during all recombinations up to the one
/// that led to an n-jet final state; identical to exclusive_dmerge, except in cases
/// where the dmin do not increase monotonically.
double exclusive_dmerge_max (const int & n) const;
```

The first returns the d_{min} in going from $n + 1$ to n jets. Occasionally however the d_{min} value does not increase monotonically during successive mergings and using a d_{cut} smaller than the return value from `exclusive_dmerge` does not guarantee an event with more than n jets. For this reason the second function `exclusive_dmerge_max` is provided — using a d_{cut} below its return value is guaranteed to provide a final state with more than n jets, while using a larger value will return a final state with n or fewer jets.

For e^+e^- collisions, where it is usual to refer to $y_{ij} = d_{ij}/Q^2$ (Q is the total (visible) energy) FastJet provides the following methods:

```
double exclusive_ymerge (int njets);
double exclusive_ymerge_max (int njets);
int n_exclusive_jets_ycut (double ycut);
std::vector<PseudoJet> exclusive_jets_ycut (double ycut);
```

which are relevant for use with the e^+e^- k_t algorithm and with the Jade plugin (section 5.4.2).

discarded) and (b) clustering is stopped when all d_{ij} and d_{iB} are above some d_{cut} . In the exclusive mode R is commonly set to 1.

The inclusive and exclusive variants are both obtained through

```
JetDefinition jet_def(kt_algorithm, R);
ClusterSequence cs(particles, jet_def);
```

The clustering sequence is identical in the inclusive and exclusive cases and the jets can then be obtained as follows:

```
vector<PseudoJet> inclusive_kt_jets = cs.inclusive_jets();
vector<PseudoJet> exclusive_kt_jets = cs.exclusive_jets(dcut);
```

4.2 Cambridge/Aachen jet algorithm

Currently the pp Cambridge/Aachen (C/A) jet algorithm [22, 23] is provided only in an inclusive version [23], whose formulation is identical to that of the k_t jet algorithm, except as regards the distance measures, which are:

$$d_{ij} = \Delta R_{ij}^2 / R^2, \quad (6a)$$

$$d_{iB} = 1. \quad (6b)$$

To use this algorithm, define

```
JetDefinition jet_def(cambridge_algorithm, R);
```

and then extract inclusive jets from the cluster sequence.

Attempting to extract exclusive jets from the Cambridge/Aachen algorithm with a d_{cut} parameter simply provides the set of jets obtained up to the point where all $d_{ij}, d_{iB} > d_{cut}$. Having clustered with some given R , this can actually be an effective way of viewing the event at a smaller radius, $R_{eff} = \sqrt{d_{cut}}R$, thus allowing a single event to be viewed at a continuous range of R_{eff} within a single clustering.

We note that the true exclusive formulation of the Cambridge algorithm [22] (in e^+e^-) instead makes use an auxiliary (k_t) distance measure and ‘freezes’ pseudojets whose recombination would involve too large a value of the auxiliary distance measure. Details are given in section 5.4.1.

4.3 Anti- k_t jet algorithm

This algorithm, introduced and studied in [14], is defined exactly like the standard k_t algorithm, except for the distance measures which are now given by

$$d_{ij} = \min(1/p_{ti}^2, 1/p_{tj}^2) \Delta R_{ij}^2 / R^2, \quad (7a)$$

$$d_{iB} = 1/p_{ti}^2. \quad (7b)$$

While it is a sequential recombination algorithm like k_t and Cambridge/Aachen, the anti- k_t algorithm behaves in some sense like a ‘perfect’ cone algorithm, in that its hard jets are exactly circular on the y - ϕ cylinder [14]. To use this algorithm, define

```
JetDefinition jet_def(antikt_algorithm, R);
```

Jet substructure as a new Higgs search channel at the LHC

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It is widely considered that, for Higgs boson searches at the Large Hadron Collider, WH and ZH production where the Higgs boson decays to $b\bar{b}$ are poor search channels due to large backgrounds. We show that at high transverse momenta, employing state-of-the-art jet reconstruction and decomposition techniques, these processes can be recovered as promising search channels for the standard model Higgs boson around 120 GeV in mass.

A key aim of the Large Hadron Collider (LHC) at CERN is to discover the Higgs boson, the particle at the heart of the standard-model (SM) electroweak symmetry breaking mechanism. Current electroweak fits, together with the LEP exclusion limit, favour a light Higgs boson, i.e. one around 120 GeV in mass [1]. This mass region is particularly challenging for the LHC experiments, and any SM Higgs-boson discovery is expected to rely on a combination of several search channels, including gluon fusion $\rightarrow H \rightarrow \gamma\gamma$, vector boson fusion, and associated production with $t\bar{t}$ pairs [2, 3].

Two significant channels that have generally been considered less promising are those of Higgs-boson production in association with a vector boson, $pp \rightarrow WH, ZH$, followed by the dominant light Higgs boson decay, to two b -tagged jets. If there were a way to recover the WH and ZH channels it could have a significant impact on Higgs boson searches at the LHC. Furthermore these two channels also provide unique information on the couplings of a light Higgs boson separately to W and Z bosons.

Reconstructing W or Z associated $H \rightarrow b\bar{b}$ production would typically involve identifying a leptonically decaying vector boson, plus two jets tagged as containing b -mesons. Two major difficulties arise in a normal search scenario. The first is related to detector acceptance: leptons and b -jets can be effectively tagged only if they are reasonably central and of sufficiently high transverse momentum. The relatively low mass of the VH (i.e. WH or ZH) system means that in practice it can be produced at rapidities somewhat beyond the acceptance, and it is also not unusual for one or more of the decay products to have too small a transverse momentum. The second issue is the presence of large backgrounds with intrinsic scales close to a light Higgs mass. For example, $t\bar{t}$ events can produce a leptonically decaying W , and in each top-quark rest frame, the b -quark has an energy of ~ 65 GeV, a value uncomfortably close to the $m_H/2$ that comes from a decaying light Higgs boson. If the second W -boson decays along the beam direction, then such a $t\bar{t}$ event can be hard to distinguish from a WH signal event.

In this letter we investigate VH production in a boosted regime, in which both bosons have large transverse momenta and are back-to-back. This region cor-

responds to only a small fraction of the total VH cross section (about 5% for $p_T > 200$ GeV), but it has several compensating advantages: (i) in terms of acceptance, the larger mass of the VH system causes it to be central, and the transversely boosted kinematics of the V and H ensures that their decay products will have sufficiently large transverse momenta to be tagged; (ii) in terms of backgrounds, it is impossible for example for an event with on-shell top-quarks to produce a high- p_T $b\bar{b}$ system and a compensating leptonically decaying W , without there also being significant additional jet activity; (iii) the HZ with $Z \rightarrow \nu\bar{\nu}$ channel becomes visible because of the large missing transverse energy.

One of the keys to successfully exploiting the boosted VH channels will lie in the use of jet-finding geared to identifying the characteristic structure of a fast-moving Higgs boson that decays to b and $\bar{b}$ in a common neighbourhood in angle. We will therefore start by describing the method we adopt for this, which builds on previous work on heavy Higgs decays to boosted W 's [4], WW scattering at high energies [5] and the analysis of SUSY decay chains [6]. We shall then proceed to discuss event generation, our precise cuts and finally show our results.

When a fast-moving Higgs boson decays, it produces a single fat jet containing two b quarks. A successful identification strategy should flexibly adapt to the fact that the $b\bar{b}$ angular separation will vary significantly with the Higgs p_T and decay orientation, roughly

$$R_{b\bar{b}} \simeq \frac{1}{\sqrt{z(1-z)}} \frac{m_H}{p_T}, \quad (p_T \gg m_H), \quad (1)$$

where $z, 1-z$ are the momentum fractions of the two quarks. In particular one should capture the $b, \bar{b}$ and any gluons they emit, while discarding as much contamination as possible from the underlying event (UE), in order to maximise resolution on the jet mass. One should also correlate the momentum structure with the directions of the two b -quarks, and provide a way of placing effective cuts on the z fractions, both of these aspects serving to eliminate backgrounds.

To flexibly resolve different angular scales we use the inclusive, longitudinally invariant Cambridge/Aachen (C/A) algorithm [7, 8]: one calculates the angular distance $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$ between all pairs of

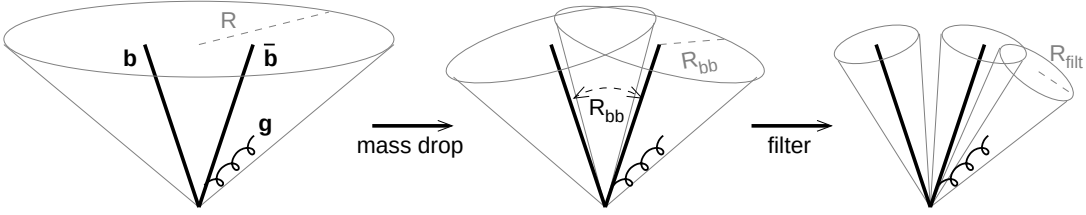


FIG. 1: The three stages of our jet analysis: starting from a hard massive jet on angular scale R , one identifies the Higgs neighbourhood within it by undoing the clustering (effectively shrinking the jet radius) until the jet splits into two subjects each with a significantly lower mass; within this region one then further reduces the radius to R_{filt} and takes the three hardest subjects, so as to filter away UE contamination while retaining hard perturbative radiation from the Higgs decay products.

objects (particles) i and j , recombines the closest pair, updates the set of distances and repeats the procedure until all objects are separated by a $\Delta R_{ij} > R$, where R is a parameter of the algorithm. It provides a hierarchical structure for the clustering, like the $K_{\perp}$ algorithm [9, 10], but in angles rather than in relative transverse momenta (both are implemented in FastJet 2.3[11]).

Given a hard jet j , obtained with some radius R , we then use the following new iterative decomposition procedure to search for a generic boosted heavy-particle decay. It involves two dimensionless parameters, μ and y_{cut} :

1. Break the jet j into two subjects by undoing its last stage of clustering. Label the two subjects j_1, j_2 such that $m_{j_1} > m_{j_2}$.
2. If there was a significant mass drop (MD), $m_{j_1} < \mu m_j$, and the splitting is not too asymmetric, $y = \frac{\min(p_{tj_1}^2, p_{tj_2}^2)}{m_j^2} \Delta R_{j_1, j_2}^2 > y_{\text{cut}}$, then deem j to be the heavy-particle neighbourhood and exit the loop. Note that $y \simeq \min(p_{tj_1}, p_{tj_2}) / \max(p_{tj_1}, p_{tj_2})$.¹
3. Otherwise redefine j to be equal to j_1 and go back to step 1.

The final jet j is to be considered as the candidate Higgs boson if both j_1 and j_2 have b tags. One can then identify $R_{b\bar{b}}$ with $\Delta R_{j_1, j_2}$. The effective size of jet j will thus be just sufficient to contain the QCD radiation from the Higgs decay, which, because of angular ordering [12, 13, 14], will almost entirely be emitted in the two angular cones of size $R_{b\bar{b}}$ around the b quarks.

The two parameters μ and y_{cut} may be chosen independently of the Higgs mass and p_T . Taking $\mu \gtrsim 1/\sqrt{3}$ ensures that if, in its rest frame, the Higgs decays to a Mercedes $b\bar{b}g$ configuration, then it will still trigger the mass drop condition (we actually take $\mu = 0.67$). The cut on $y \simeq \min(z_{j_1}, z_{j_2}) / \max(z_{j_1}, z_{j_2})$ eliminates the asymmetric configurations that most commonly generate significant jet masses in non- b or single- b jets, due to the

Jet definition	σ_S/fb	σ_B/fb	$S/\sqrt{B} \cdot \text{fb}$
C/A, $R = 1.2$, MD-F	0.57	0.51	0.80
$K_{\perp}$, $R = 1.0$, y_{cut}	0.19	0.74	0.22
SISCone, $R = 0.8$	0.49	1.33	0.42

TABLE I: Cross section for signal and the Z +jets background in the leptonic Z channel for $200 < p_{TZ}/\text{GeV} < 600$ and $110 < m_J/\text{GeV} < 125$, with perfect b -tagging; shown for our jet definition, and other standard ones at near optimal R values.

soft gluon divergence. It can be shown that the maximum $S/\sqrt{B}$ for a Higgs boson compared to mistagged light jets is to be obtained with $y_{\text{cut}} \simeq 0.15$. Since we have mixed tagged and mistagged backgrounds, we use a slightly smaller value, $y_{\text{cut}} = 0.09$.

In practice the above procedure is not yet optimal for LHC at the transverse momenta of interest, $p_T \sim 200 - 300 \text{ GeV}$ because, from eq. (1), $R_{b\bar{b}} \gtrsim 2m_H/p_T$ is still quite large and the resulting Higgs mass peak is subject to significant degradation from the underlying event (UE), which scales as $R_{b\bar{b}}^4$ [15]. A second novel element of our analysis is to *filter* the Higgs neighbourhood. This involves resolving it on a finer angular scale, $R_{\text{filt}} < R_{b\bar{b}}$, and taking the three hardest objects (subjects) that appear — thus one captures the dominant $\mathcal{O}(\alpha_s)$ radiation from the Higgs decay, while eliminating much of the UE contamination. We find $R_{\text{filt}} = \min(0.3, R_{b\bar{b}}/2)$ to be rather effective. We also require the two hardest of the subjects to have the b tags.

The overall procedure is sketched in Fig. 1. We illustrate its effectiveness by showing in table I (a) the cross section for identified Higgs decays in HZ production, with $m_H = 115 \text{ GeV}$ and a reconstructed mass required to be in an moderately narrow (but experimentally realistic) mass window, and (b) the cross section for background $Zb\bar{b}$ events in the same mass window. Our results (C/A MD-F) are compared to those for the $K_{\perp}$ algorithm with the same y_{cut} and the SISCone [16] algorithm based just on the jet mass. The $K_{\perp}$ algorithm does well on background rejection, but suffers in mass resolution, leading to a low signal; SISCone takes in less UE so gives good resolution on the signal, however, because it ignores the underlying substructure, fares poorly on background rejection. C/A MD-F performs well both

¹ Note also that this y_{cut} is related to, but not the same as, that used to calculate the splitting scale in [5, 6], which takes the jet p_T as the reference scale rather than the jet mass.

on mass resolution and background rejection.

The above results were obtained with HERWIG 6.510[17, 18] with JIMMY 4.31 [19] for the underlying event, which has been used throughout the subsequent analysis. The signal reconstruction was also cross-checked using PYTHIA 6.403[20]. In both cases the underlying event model was chosen in line with the tunes currently used by ATLAS and CMS (see for example [21]²). The leading-logarithmic parton shower approximation used in these programs have been shown to model jet substructure well in a wide variety of processes [23, 24, 25, 26, 27, 28]. For this analysis, signal samples of WH, ZH were generated, as well as $WW, ZW, ZZ, Z + \text{jet}, W + \text{jet}, t\bar{t}$, single top and dijets to study backgrounds. All samples correspond to a luminosity $\geq 30 \text{ fb}^{-1}$, except for the lowest $\hat{p}_T^{\text{min}}$ dijet sample, where the cross section makes this impractical. In this case an assumption was made that the selection efficiency of a leptonically-decaying boson factorises from the hadronic Higgs selection. This assumption was tested and is a good approximation in the signal region of the mass plot, though correlations are significant at lower masses.

The leading order (LO) estimates of the cross-section were checked by comparing to next-to-leading order (NLO) results. High- p_T VH and $Vb\bar{b}$ cross sections were obtained with MCFM [29, 30] and found to be about 1.5 times the LO values for the two signal and the $Z^0b\bar{b}$ channels (confirmed with MC@NLO v3.3 for the signal [31]), while the $W^\pm b\bar{b}$ channel has a K-factor closer to 2.5 (as observed also at low- p_T in [30]).³ The main other background, $t\bar{t}$ production, has a K-factor of about 2 (found comparing the HERWIG total cross section to [32]). This suggests that our final LO-based signal/ $\sqrt{\text{background}}$ estimates ought not to be too strongly affected by higher order corrections, though further detailed NLO studies would be of value.

Let us now turn to the details of the event selection. The candidate Higgs jet should have a p_T greater than some $\hat{p}_T^{\text{min}}$. The jet R -parameter values commonly used by the experiments are typically in the range 0.4 - 0.7. Increasing the R -parameter increases the fraction of contained Higgs decays. Scanning the region $0.6 < R < 1.6$ for various values of $\hat{p}_T^{\text{min}}$ indicates an optimum value around $R = 1.2$ with $\hat{p}_T^{\text{min}} = 200 \text{ GeV}$.

Three subselections are used for vector bosons: (a) An e^+e^- or $\mu^+\mu^-$ pair with an invariant mass $80 \text{ GeV} < m < 100 \text{ GeV}$ and $p_T > \hat{p}_T^{\text{min}}$. (b) Missing transverse momentum $> \hat{p}_T^{\text{min}}$. (c) Missing transverse momentum

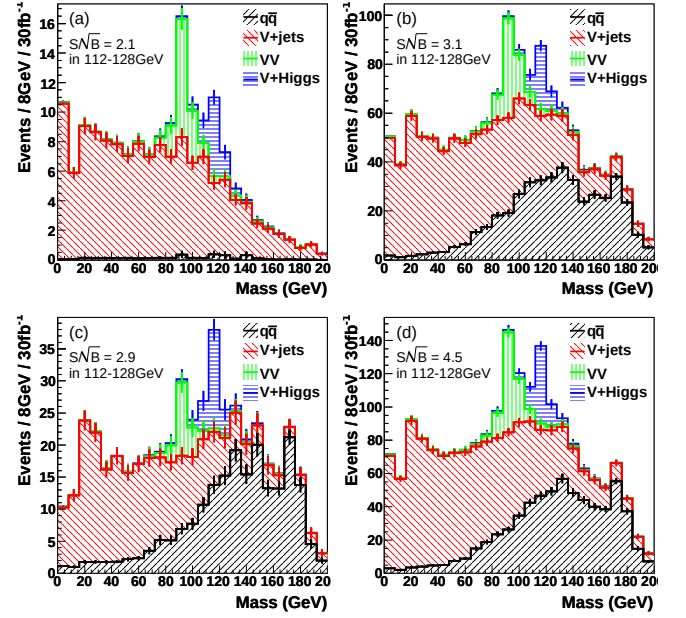


FIG. 2: Signal and background for a 115 GeV SM Higgs simulated using HERWIG, C/A MD-F with $R = 1.2$ and $p_T > 200 \text{ GeV}$, for 30 fb^{-1} . The b tag efficiency is assumed to be 60% and a mistag probability of 2% is used. The $q\bar{q}$ sample includes dijets and $t\bar{t}$. The vector boson selections for (a), (b) and (c) are described in the text, and (d) shows the sum of all three channels. The errors reflect the statistical uncertainty on the simulated samples, and correspond to integrated luminosities $> 30 \text{ fb}^{-1}$.

$> 30 \text{ GeV}$ plus a lepton (e or μ) with $p_T > 30 \text{ GeV}$, consistent with a W of nominal mass with $p_T > \hat{p}_T^{\text{min}}$. It may also be possible, by using similar techniques to reconstruct hadronically decaying bosons, to recover signal from these events. This is a topic left for future study.

To reject backgrounds we require that there be no leptons with $|\eta| < 2.5, p_T > 30 \text{ GeV}$ apart from those used to reconstruct the leptonic vector boson, and no b -tagged jets in the range $|\eta| < 2.5, p_T > 50 \text{ GeV}$ apart from the Higgs candidate. For channel (c), where the $t\bar{t}$ background is particularly severe, we require that there are no additional jets with $|\eta| < 3, p_T > 30 \text{ GeV}$. The rejection might be improved if this cut were replaced by a specific top veto [5]. However, without applying the sub-jet mass reconstruction to all jets, the mass resolution for $R = 1.2$ is inadequate.

The results for $R = 1.2, \hat{p}_T^{\text{min}} = 200 \text{ GeV}$ are shown in Fig. 2, for $m_H = 115 \text{ GeV}$. The Z peak from ZZ and WZ events is clearly visible in the background, providing a critical calibration tool. Relaxing the b -tagging selection would provide greater statistics for this calibration, and would also make the W peak visible. The major backgrounds are from W or Z +jets, and (except for the $HZ(Z \rightarrow l^+l^-)$ case), $t\bar{t}$.

Combining the three sub-channels in Fig. 2d, and summing signal and background over the two bins in the range 112-128 GeV, the Higgs is seen with a significance

² The non-default parameter setting are: PRSOF=0, JMRAD(73)=1.8, PTJIM=4.9 GeV, JMUEO=1, with CTEQ6L [22] PDFs.

³ For the $Vb\bar{b}$ backgrounds these results hold as long as both the vector boson and $b\bar{b}$ jet have a high p_T ; relaxing the requirement on p_{TV} leads to enhanced K -factors from electroweak double-logarithms.

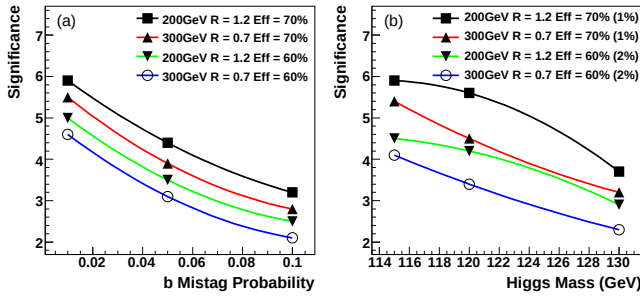


FIG. 3: Estimated sensitivity for 30 fb^{-1} under various different sets of cuts and assumptions (a) for $m_H = 115 \text{ GeV}$ as a function of the mistag probability for b -subjects and (b) as a function of Higgs mass for the b -tag efficiency (mistag rates) shown in the legend. Significance is estimated as $\text{signal}/\sqrt{\text{background}}$ in the peak region.

of 4.5σ (8.2σ for 100 fb^{-1}). The intrinsic resolution of the jet mass at the particle level would allow finer binning and greater significance. However, studies [33, 34] using parameterised simulations of the ATLAS detector indicate that detector resolution would prohibit this.

The b -tagging and mistag probabilities are critical parameters for this analysis, and no detailed study has been published of tagging two high- p_T b subjects. Values used by experiments for single-tag probabilities range up to 70% for the efficiency and down to 1% for mistags. Results for 70% and 60% efficiency are summarised in Fig. 3a as a function of the mistag probability.

There is a trade-off between rising cross-section and falling fraction of contained decays (as well as rising backgrounds) as $\hat{p}_T^{\min}$ is reduced. As an example of the dependence on this trade-off, we show the sensitivity for $\hat{p}_T^{\min} = 300 \text{ GeV}$, $R = 0.7$ in Fig. 3a.

The significance falls for higher Higgs masses, as shown in Fig. 3b, but values of 3σ or above seem achievable up to $m_H = 130 \text{ GeV}$.

In addition to the b -tagging, the effects of pile-up, intrinsic resolution and granularity of the detector will all have an impact. Several ideas exist to improve some of these, and initial studies with realistic detector simulations indicate that the efficiencies and resolutions assumed here are not unreasonable, though the exact requirements of our analysis have not been studied with such tools.

We conclude that subjet techniques have the potential to transform the high- p_T $WH, ZH(H \rightarrow b\bar{b})$ channel into one of the best channels for discovery of a low mass Standard Model Higgs at the LHC. This channel could also provide unique information on the coupling of the Higgs boson separately to W and Z bosons. Realising this potential is a challenge that merits further experimental study and complementary theoretical investigations.

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before, this precise picture holds for our specific choice of z_{cut} .

The position of the peak for the I component, in the case of quark-initiated jets, is in reasonable agreement with the one determined by the fixed-coupling approximation, Eq. (5.13), though the agreement is poorer for gluon jets: for a reliable quantitative treatment of the peak region it is important to include subleading terms.

6 Mass Drop Tagger

The mass-drop tagger [23] was designed to be used with jets found by the Cambridge/Aachen algorithm [39, 40]. It involves two parameters y_{cut} and μ and, for an initial jet labelled j , proceeds as follows:

1. Break the jet j into two subjets by undoing its last stage of clustering. Label the two subjets j_1, j_2 such that $m_{j_1} > m_{j_2}$.
2. If there was a significant mass drop, $m_{j_1} < \mu m_j$, and the splitting is not too asymmetric, $y = \min(p_{tj_1}^2, p_{tj_2}^2) \Delta R_{j_1 j_2}^2 / m_j^2 > y_{\text{cut}}$, then deem j to be the tagged jet.
3. Otherwise redefine j to be equal to j_1 and go back to step 1 (unless j consists of just a single particle, in which case the original jet is deemed untagged).

Typical parameter choices are for example $\mu = 2/3$ and y_{cut} in the range $0.09 - 0.15$. While the y_{cut} parameter will appear explicitly in our results, μ will not, and indeed we shall see that its exact value is not critical as long as it is not parametrically small.

6.1 Leading order calculation

As usual, it is useful to start with a leading-order configuration, for which the jet consists of just two partons. When the jet is declustered, each of the prongs is massless, so that the mass-drop condition is automatically satisfied, rendering the μ parameter irrelevant. There are then two possibilities: if the asymmetry condition is satisfied the jet is tagged, with the tagged mass equal to the original jet mass. Otherwise the jet does not contribute to the tagged jet mass distribution.

Considering a quark that splits into a quark with momentum fraction $1 - z$ and a gluon with momentum fraction z , we have $m_j^2 = z(1 - z)E^2$. The asymmetry condition then becomes $\frac{z}{1 - z} > y_{\text{cut}}$ and $\frac{1 - z}{z} > y_{\text{cut}}$.

We may now write the differential cross section for the jet to have a given tagged mass:

$$\begin{aligned} \frac{1}{\sigma} \frac{d\sigma}{dm^2}^{(\text{MDT, LO})} &= C_F \frac{\alpha_s}{\pi} \int dz p_{gq}(z) \frac{d\theta^2}{\theta^2} \delta(m^2 - z(1 - z)p_t^2 \theta^2) \times \\ &\quad \times \Theta\left(\frac{z}{1 - z} - y_{\text{cut}}\right) \Theta\left(\frac{1 - z}{z} - y_{\text{cut}}\right) \Theta(R^2 - \theta^2). \end{aligned} \quad (6.1)$$

Proceeding as with our other LO calculations, including a requirement $y_{\text{cut}} \ll 1$, leads us to the following result

$$\frac{\rho}{\sigma} \frac{d\sigma}{d\rho}^{(\text{MDT, LO})} = \frac{\alpha_s C_F}{\pi} \left[\Theta(\rho - y_{\text{cut}}) \ln \frac{1}{\rho} + \Theta(y_{\text{cut}} - \rho) \ln \frac{1}{y_{\text{cut}}} - \frac{3}{4} \right]. \quad (6.2)$$

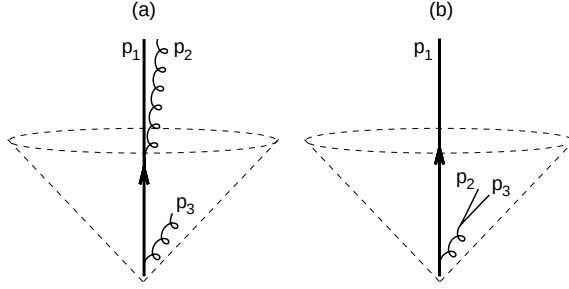


Figure 8. Two characteristic partonic configurations that arise at in the tree-level $\mathcal{O}(\alpha_s^2)$ contribution. The dashed cone provides a schematic representation of the boundary of the jet.

Modulo the replacement $z_{\text{cut}} \rightarrow y_{\text{cut}}$, this is identical to the result for pruning, Eq. (5.3), and in particular has two regimes: it is linear in $\ln \frac{1}{\rho}$ when $\rho > y_{\text{cut}}$, and saturates at a constant value ($\ln \frac{1}{y_{\text{cut}}} - \frac{3}{4}$) for $\rho < y_{\text{cut}}$. In contrast to the case of pruning, it is intriguing that this structure appears rather similar to what is observed in the Monte Carlo results for quark jets in Fig. 1. This would suggest that there are cases where effects beyond LO might be modest.

6.2 3-particle configurations

The next step in understanding the mass-drop tagger is to consider 3-particle configurations, where for the first time one encounters the recursive nature of the tagger and potentially also the dependence on μ .

Since we will be mainly interested in logarithmically enhanced contributions, we can exploit the fact that these come from configurations in which momenta are ordered in angle and/or energy. Some interesting such configurations are illustrated in Fig. 8.

Configuration (a) has the ordering $\theta_{13} \gg \theta_{12}$, with the ordering sufficiently strong that we can assume $m_{\text{jet}} = m_{123} \gg m_{12}$. Because the jet was clustered with the angular-ordered C/A algorithm, the MDT first splits the jet into j_{12} and j_3 . If $E_3/E_{12} > y_{\text{cut}}$ then the declustering passes the asymmetry cut; the strong angular ordering ensures that it also passes the mass-drop condition and so the jet as a whole is tagged. If $E_3/E_{12} < y_{\text{cut}}$, then the MDT recurses, into the heavier of the two subjets, i.e. j_{12} , which can be analysed as in the previous, LO section. The key point here is that in the limit in which $E_3 \ll E_{\text{jet}}$, the presence of gluon 3 has no effect on whether the j_{12} system gets tagged. This is true even though we chose a configuration where m_{jet} is dominated by emission 3. This was part of the intended design of the MDT: if the jet contains hard substructure, the tagger should find it, even if there is other soft structure (including underlying event and pileup) that strongly affects the original jet mass. It is possible to show that if one combines the NLO contribution that comes from configurations like (a) with the corresponding virtual graphs, one obtains a contribution to $\Sigma^{(\text{MDT})}(\rho)$ that goes as $\alpha_s^2 L^2$ for arbitrarily large L . This involves fewer logarithms than any of the plain jet mass, trimming or pruning. However it turns out not to be the leading contribution in terms of a counting of logarithms and therefore we postpone its detailed discussion.

Configuration (b) in Fig. 8 reveals an unintended behaviour of the tagger. Here we have $\theta_{23} \ll \theta_{12} \simeq \theta_{13}$, so the first unclustering leads to j_1 and j_{23} subjets. It may happen that the parent gluon of the j_{23} subjet was soft, so that $E_{23} < y_{\text{cut}} E_{\text{jet}}$. The jet therefore fails the symmetry requirement at this stage, and so recurses one step down. The formulation of the MDT is such that it recurses into the more massive of the two prongs, i.e. only follows the j_{23} prong, even though this is soft. This was not what was intended in the original design, and is to be considered a flaw — in essence one follows the wrong branch.

It is interesting to determine the logarithmic structure that results from the wrong-branch issue. Exceptionally, we are going to work in an approximation in which we treat logarithms of y_{cut} on the same footing as logarithms of ρ . We will, however, neglect terms that do not have the maximal number of logarithms of either argument. The wrong-branch distribution can then be written as

$$\begin{aligned} \frac{\rho}{\sigma} \frac{d\sigma}{d\rho}^{(\text{MDT}, \text{NLO}_{\text{flaw}})} &= \rho C_F C_A \left(\frac{\alpha_s}{\pi} \right)^2 \int \frac{dx}{x} \frac{d\theta^2}{\theta^2} \Theta(R^2 - \theta^2) \Theta(y_{\text{cut}} - x) \times \\ &\times \int^1 \frac{dz}{z} \Theta(z - y_{\text{cut}}) \int \frac{d\theta_{23}^2}{\theta_{23}^2} \delta\left(\rho - zx^2 \frac{\theta_{23}^2}{R^2}\right) \Theta(\theta - \theta_{23}) \end{aligned} \quad (6.3)$$

where θ is the angle between j_1 and the j_{23} system, while $x = E_{23}/E_{\text{jet}}$ and $z = E_2/E_{23}$. In writing the constraints on the angles, we have assumed strong-ordering of the angles. We are also working in a soft approximation, $x \ll 1$ and $z \ll 1$. The answer is non-zero only for $\rho \lesssim y_{\text{cut}}^2$, because x must be less than y_{cut} , while the maximum θ_{23} angle is of order R^2 .¹¹ If $\rho \gtrsim y_{\text{cut}}^3$ then the y_{cut} condition in the second line of Eq. (6.3) does not play a role, and one obtains

$$\frac{\rho}{\sigma} \frac{d\sigma}{d\rho}^{(\text{MDT}, \text{NLO}_{\text{flaw}})} = \frac{C_F C_A}{12} \left(\frac{\alpha_s}{\pi} \right)^2 \ln^3 \frac{y_{\text{cut}}^2}{\rho}, \quad y_{\text{cut}}^3 \lesssim \rho \lesssim y_{\text{cut}}^2, \quad (6.4)$$

otherwise the result is

$$\begin{aligned} \frac{\rho}{\sigma} \frac{d\sigma}{d\rho}^{(\text{MDT}, \text{NLO}_{\text{flaw}})} &= \frac{C_F C_A}{4} \left(\frac{\alpha_s}{\pi} \right)^2 \left(\ln^2 \frac{y_{\text{cut}}^3}{\rho} \ln \frac{1}{y_{\text{cut}}} + \right. \\ &\left. + \ln \frac{y_{\text{cut}}^2}{\rho} \ln^2 \frac{1}{y_{\text{cut}}} - \frac{2}{3} \ln^3 \frac{1}{y_{\text{cut}}} \right), \quad \rho \lesssim y_{\text{cut}}^3. \end{aligned} \quad (6.5)$$

Considering just the asymptotically small- ρ region, which starts for $\rho \lesssim y_{\text{cut}}^3$, the integrated distribution, $\Sigma^{(\text{MDT})}(\rho)$ has a logarithmic structure $\alpha_s^2 L^3 \ln \frac{1}{y_{\text{cut}}}$, i.e. enhanced by $\alpha_s L^2$ relative to the LO result and by a power of $L/\ln \frac{1}{y_{\text{cut}}}$ relative to configurations of type (a).

Based on the above calculation, one might expect the “wrong-branch” contributions to dominate over the LO type behaviour. In practice they don’t. Part of the reason for this

¹¹In the phase-space region where $\theta \sim \theta_{23} \sim R$, the approximation of strongly ordered angles is inappropriate. The determination of the exact onset of the wrong-branch issue would require a full treatment of that region. One would also need to go beyond the small- z approximation: insofar as the squared jet mass involves a factor $z(1-z)$ rather than simply z , one would then expect an onset in the neighbourhood of $\rho \sim y_{\text{cut}}^2/4$ rather than y_{cut}^2 . However, in terms of a logarithmic counting, these considerations should only affect subleading logarithms.

1 Introduction

The study of jet substructure has significantly matured over the past five years [1–3], with numerous techniques proposed to tag boosted objects [4–46], distinguish quark from gluon jets [44, 47–51], and mitigate the effects of jet contamination [6, 52–61]. Many of these techniques have found successful applications in jet studies at the Large Hadron Collider (LHC) [50, 62–89], and jet substructure is likely to become even more relevant with the anticipated increase in energy and luminosity for Run II of the LHC.

In addition to these phenomenological and experimental studies of jet substructure, there is a growing catalog of first-principles calculations using perturbative QCD (pQCD). These include more traditional jet mass and jet shape distributions [90–95] as well as more sophisticated substructure techniques [44, 59, 60, 96–103]. Recently, Refs. [59, 60] considered the analytic behavior of three of the most commonly used jet tagging/grooming methods—trimming [53], pruning [54, 55], and mass drop tagging [6]. Focusing on groomed jet mass distributions, this study showed how their qualitative and quantitative features could be understood with the help of logarithmic resummation. Armed with this analytic understanding of jet substructure, the authors of Ref. [59] developed the modified mass drop tagger (mMDT) which exhibits some surprising features in the resulting groomed jet mass distribution, including the absence of Sudakov double logarithms, the absence of non-global logarithms [104], and a high degree of insensitivity to non-perturbative effects.

In this paper, we introduce a new tagging/grooming method called “soft drop declustering”, with the aim of generalizing (and in some sense simplifying) the mMDT procedure. Like any grooming method, soft drop declustering removes wide-angle soft radiation from a jet in order to mitigate the effects of contamination from initial state radiation (ISR), underlying event (UE), and multiple hadron scattering (pileup). Given a jet of radius R_0 with only two constituents, the soft drop procedure removes the softer constituent unless

$$\text{Soft Drop Condition: } \frac{\min(p_{T1}, p_{T2})}{p_{T1} + p_{T2}} > z_{\text{cut}} \left(\frac{\Delta R_{12}}{R_0} \right)^\beta, \quad (1.1)$$

where p_{Ti} are the transverse momenta of the constituents with respect to the beam, ΔR_{12} is their distance in the rapidity-azimuth plane, z_{cut} is the soft drop threshold, and β is an angular exponent. By construction, Eq. (1.1) fails for wide-angle soft radiation. The degree of jet grooming is controlled by z_{cut} and β , with $\beta \rightarrow \infty$ returning back an ungroomed jet. As we explain in Sec. 2, this procedure can be extended to jets with more than two constituents with the help of recursive pairwise declustering.¹

Following the spirit of Ref. [59], the goal of this paper is to understand the analytic behavior of the soft drop procedure, particularly as the angular exponent β is varied. There are two different regimes of interest. For $\beta > 0$, soft drop declustering removes soft radiation

¹The soft drop procedure takes some inspiration from the “semi-classical jet algorithm” [58], where a variant of Eq. (1.1) with $z_{\text{cut}} = 1/2$ and $\beta = 3/2$ is tested at each stage of recursive clustering (unlike declustering considered here).

however, a few important differences. First, soft drop declustering does not require a mass drop condition (or equivalently, the mass drop parameter μ is set to unity). As shown in Ref. [59], the mass drop condition is largely irrelevant for understanding the analytic behavior of mMDT on quark/gluon jets, so we have decided not to include it in the definition here. Second, we note that the $\beta = 0$ limit corresponds to a mMDT variant where step 2 is implemented directly on the transverse momentum fractions of subjets, rather than indirectly through a ratio of a k_t -distance to a mass [59]. Of course, the two give the same behavior in the small $\min(p_{T1}, p_{T2})/(p_{T1} + p_{T2})$ limit, but Eq. (1.1) makes it obvious that the soft drop condition drops soft radiation (true to its name). Finally and most importantly, for $\beta \neq 0$, the soft drop condition involves a relation between energies and angular distances, rather than just energies as is the case for $\beta = 0$. It is this additional angular dependence (exploited by the exponent β) that we wish to highlight in this paper.

As mentioned in footnote 1, the soft drop condition takes some inspiration from the “semi-classical jet algorithm” [58]. The semi-classical algorithm is a pairwise clustering algorithm that only allows mergings which satisfy

$$\text{Semi-classical Condition: } \frac{\min(m_{T1}, m_{T2})}{m_{T1} + m_{T2}} > \frac{1}{2} \left(\frac{\Delta R_{12}}{R_0} \right)^{3/2}, \quad (2.1)$$

where $m_{Ti} = \sqrt{m_i^2 + p_{Ti}^2}$. Apart from the change of $p_{Ti} \rightarrow m_{Ti}$, the semi-classical condition looks like the soft drop condition with $\beta = 3/2$ and $z_{\text{cut}} = 1/2$, but there is an important difference. For semi-classical jets, one is recursively clustering a jet using a novel measure. For soft-drop jets, one is taking an existing jet defined with a traditional algorithm and using soft drop declustering to groom away soft wide-angle emissions. Of course, the distinction between clustering and declustering is irrelevant for a jet with only two constituents, but it is very important for our analytic calculations which only apply to declustering of a C/A tree.³

2.2 Dependence on β

Before studying the analytic behavior of soft-drop distributions in detail, it is worth making a few general comments about the expected β dependence. For simplicity of discussion, we will work with central jets (i.e. rapidity $y = 0$) with small radius ($R_0 \ll 1$). This way, we can freely exchange transverse momentum p_T for energy E , as well as rapidity-azimuth distance R for opening angle θ . All of the results of this paper extend to non-zero rapidity as well, up to power corrections in the jet radius, which we neglect.

In Fig. 1, we show the phase space for a single gluon emission from an eikonal hard quark/gluon on the $(\log \frac{1}{z}, \log \frac{R_0}{\theta})$ plane, where $0 \leq z \leq 1$ is the energy fraction and $0 \leq \theta \leq R_0$ is the angle of the emission. We have labeled three modes in the phase space: soft, soft-collinear, and collinear. For this paper, we define the various modes in terms of their z

³In principle, it is possible to use any of the generalized k_t algorithms [112, 113] to perform the soft drop declustering. The choice of C/A is motivated by the approximate angular ordering of emissions in the parton shower.

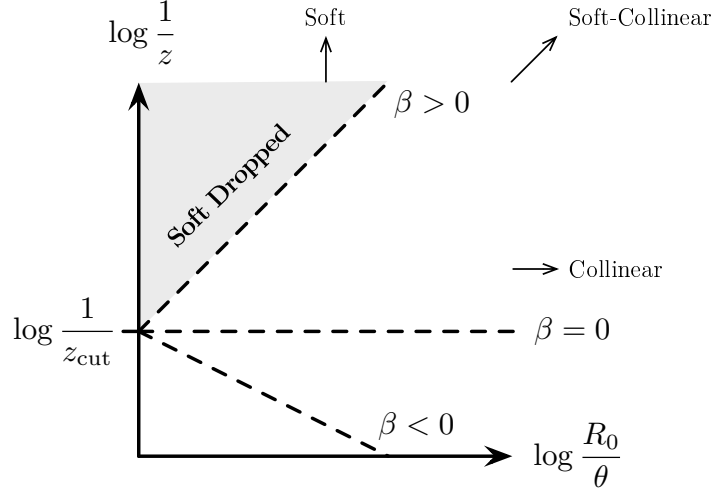


Figure 1: Phase space for emissions on the $(\log \frac{1}{z}, \log \frac{R_0}{\theta})$ plane. In the strongly-ordered limit, emissions above the dashed line (Eq. (2.2)) are vetoed by the soft drop condition. For $\beta > 0$, soft emissions are vetoed while much of the soft-collinear region is maintained. For $\beta = 0$ (mMDT), both soft and soft-collinear emissions are vetoed. For $\beta < 0$, all (two-prong) singularities are regulated by the soft drop procedure.

and θ behavior:

$$\begin{aligned} \text{soft modes: } & z \rightarrow 0, \theta = \text{constant}, \\ \text{soft-collinear modes: } & z \rightarrow 0, \theta \rightarrow 0, \\ \text{collinear modes: } & z = \text{constant}, \theta \rightarrow 0. \end{aligned}$$

No relative scaling is assumed between energy fraction z and splitting angle θ for soft-collinear modes. In these logarithmic coordinates, the emission probability is flat in the soft-collinear limit. In the soft limit, the soft drop criteria reduces to

$$z > z_{\text{cut}} \left(\frac{\theta}{R_0} \right)^\beta \quad \Rightarrow \quad \log \frac{1}{z} < \log \frac{1}{z_{\text{cut}}} + \beta \log \frac{R_0}{\theta}. \quad (2.2)$$

Thus, vetoed emissions lie above a straight line of slope β on the $(\log \frac{1}{z}, \log \frac{R_0}{\theta})$ plane, as shown in Fig. 1.

For $\beta > 0$, collinear radiation always satisfies the soft drop condition, so a soft-drop jet still contains all of its collinear radiation. The amount of soft-collinear radiation that satisfies the soft drop condition depends on the relative scaling of the energy fraction z to the angle θ . As $\beta \rightarrow 0$, more of the soft-collinear radiation of the jet is removed, and in the $\beta = 0$ (mMDT) limit, all soft-collinear radiation is removed. Therefore, we expect that the coefficient of the double logarithms of observables like groomed jet mass (and $C_1^{(\alpha)}$) will be proportional to β ,

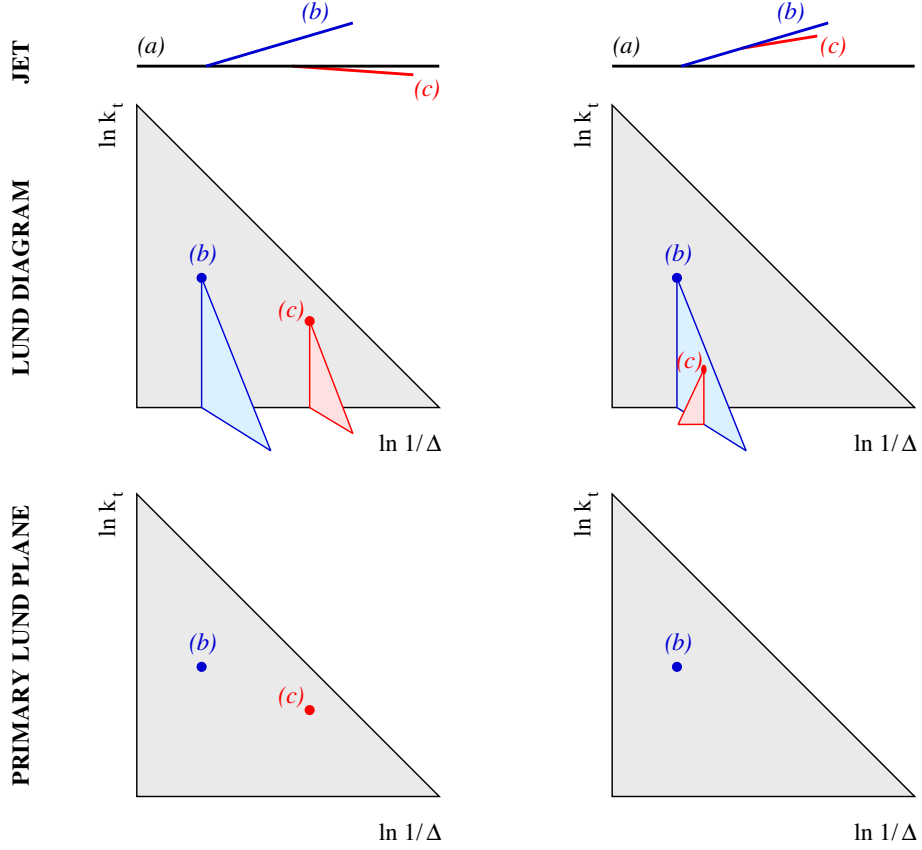


Figure 1: Different representations for two jets. Top: the particles inside the jet. Middle: the full Lund diagram. Bottom: the primary Lund plane. See text for further details.

of the corresponding particle. The black particle (a) is the primary particle, i.e. the one that initiated the jet. Particles (b) and (c) are emissions inside the jet.

The middle representation gives the full Lund diagrams for each of the two jets. The phase-space for emission from each particle is represented as a triangle in a $\ln \Delta$ and $\ln k_t$ plane, where Δ and k_t are respectively the angle and transverse momentum of an emission with respect to its emitter. The triangles are colour-coded to match the colours of the particles in the upper row. The black triangle represents the primary phase space, i.e. emission from (a) (our classification of which particle emits which other ones is based on the concept of angular ordering of emissions). Considering the left-hand jet, the blue particle (b) in the jet is represented as a blue point at the appropriate (Δ, k_t) coordinate on the (black) triangle associated with its emitter (a). The blue particle has its own phase-space region, the blue triangle, which is known as a secondary Lund triangle, or “leaf” where the particle could have, but in this case didn’t, emit. Similarly for the red particle, (c), which is also emitted from (a). In contrast, for the right-hand jet, (c) was emitted from (b) and so its point appears on the (secondary) blue triangle associated with particle (b), while its red phase-space triangle emerges as a tertiary triangle, or leaf, off (b)’s triangle.

Finally, the bottom diagram shows the primary Lund plane, which contains just the

positions of the emissions from (a), but no information about what further secondary emissions may have been produced. It is this simpler representation that we will use throughout most of the article.

2.1 Construction of the primary Lund plane

Our starting point for constructing the primary Lund plane is to (re-)cluster a jet’s constituents with the Cambridge-Aachen (C/A) algorithm [49, 50], which has significant advantages over other members of the generalised- k_t [51] family (see section 2.4).² The C/A algorithm identifies the pair of particles i and j closest in rapidity ($y = \ln \frac{E+p_z}{E-p_z}$, with E and p_z the particle’s energy and longitudinal momentum with respect to the colliding beams) and azimuth ϕ , i.e. with the minimal value of $\Delta_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$. It then recombines them into a “pseudojet” with momentum $p = p_i + p_j$. This procedure is repeated until all particles (and pseudojets) have been recombined, or are separated by Δ_{ij} larger than some parameter R .

To create a primary Lund plane representation of a jet we then work backwards through the C/A clustering. One starts with the full jet and then proceeds as follows:

1. Decluster the current object to produce two pseudojets, p_a and p_b , labelled such that $p_{ta} > p_{tb}$, where p_{ti} is the transverse momentum of i with respect to the colliding beams. We will consider p_b to be the emission and $p_a + p_b$ to be the emitter. In the limit where p_b carries little momentum relative to p_a , $p_a + p_b$ and p_a can be thought of being the same particle, simply differing through the loss of a small amount of momentum by the radiation of a gluon p_b .
2. Determine a number of variables associated with the declustering, e.g.

$$\Delta \equiv \Delta_{ab}, \quad k_t \equiv p_{tb} \Delta_{ab}, \quad m^2 \equiv (p_a + p_b)^2, \quad (2.1a)$$

$$z \equiv \frac{p_{tb}}{p_{ta} + p_{tb}}, \quad \kappa \equiv z \Delta, \quad \psi \equiv \tan^{-1} \frac{y_b - y_a}{\phi_b - \phi_a}, \quad (2.1b)$$

In the limit $p_{tb} \ll p_{ta}$ and $\Delta \ll 1$, k_t is the transverse momentum of particle b (the emission) relative to its emitter, ψ is an azimuthal angle around the (sub)jet axis, and z is the momentum fraction of the branching. In our default definition of the Lund plane, the coordinates associated with this declustering will be $\ln \Delta$ and $\ln k_t$. One may also, however, make other choices of coordinates, such as for example $\ln \Delta$ and $\ln \kappa$, or $\ln \Delta$ and $\ln k_t/p_{t,\text{jet}}$ (with $p_{t,\text{jet}}$ the jet transverse momentum). We will denote the variables as a tuple $\mathcal{T}^{(i)} = \{k_t^{(i)}, \Delta^{(i)}, \dots\}$ for the i^{th} iteration of this step.

3. Repeat the procedure by going to step 1 for the harder branch, p_a .

This procedure gives an ordered list of tuples of variables

$$\mathcal{L}_{\text{primary}} = \left[\mathcal{T}^{(1)}, \dots, \mathcal{T}^{(n)} \right] \quad (2.2)$$

²Throughout this paper, we also use the C/A algorithm for the initial jet finding. The case where jets are clustered with the anti- k_t algorithm (and re-clustered with the C/A algorithm) is discussed in Appendix A.

containing the kinematic variables for each of the primary branchings off the main emitter. The k_t and Δ elements of the tuples (specifically their logarithms) can be interpreted as set of coordinates of points in the Lund plane, corresponding to the full set of primary branchings, as in the lower row of Fig. 1. The tuple elements other than k_t and Δ provide complementary information for each point.

One could additionally follow the lower p_t branch at each declustering. This would effectively create secondary, tertiary, etc., Lund planes (or triangles), i.e. one for each emission, giving the full Lund diagram as in the middle row of Fig. 1. We postpone the study of full Lund diagrams to future work, although a brief discussion of the use of a secondary Lund plane is given in appendix B.

A last point in this subsection is to consider infrared and collinear safety. The full list of tuples produced by the primary declustering procedure is not infrared and collinear safe. For example, if one adds an infinitesimally soft emission, it may lead to an additional primary declustering, and so an extra tuple in the list. However if one considers the subset of tuples in some specified finite region of the Lund plane (for example all tuples with k_t larger than some cut, or all tuples in a given two-dimension bin of k_t and Δ), then one recovers infrared and collinear safety. Specifically, the extra soft emission will not modify that subset, because either it will be declustered as a primary emission that is outside that finite region, or it will be clustered with a harder emission inside that region, but will not modify the kinematic variables of the tuple associated with that harder emission's own primary declustering. In practice only the pattern of declusterings with $k_t \gg \Lambda_{\text{QCD}}$ is amenable to perturbative calculation.

2.2 Averaged Lund plane density and basic analytical properties

The simplest analysis of the Lund plane is to examine the average density of points per jet and per unit area in the $\ln k_t - \ln \Delta$ plane, which we denote

$$\rho(\Delta, k_t) = \frac{1}{N_{\text{jet}}} \frac{dn_{\text{emission}}}{d \ln k_t d \ln 1/\Delta}. \quad (2.3)$$

One can also define a density in terms of dimensionless variables, e.g.

$$\bar{\rho}(\Delta, \kappa) = \frac{1}{N_{\text{jet}}} \frac{dn_{\text{emission}}}{d \ln \kappa d \ln 1/\Delta}. \quad (2.4)$$

The quantity $\rho(\Delta, k_t)$ is represented in Fig. 2a for a sample of (C/A , $R = 1$) jets with $p_t > 2$ TeV, simulated using the dijet process in Pythia 8.230 [52] with the Monash13 tune [53]. For the case of a quark-initiated jet (about 80% of the jets in the sample Fig. 2a), to leading order in perturbative QCD and for $\Delta \ll 1$, one expects

$$\rho \simeq \frac{\alpha_s(k_t) C_F}{\pi} \bar{z} (p_{gq}(\bar{z}) + p_{gq}(1 - \bar{z})), \quad \bar{z} = \frac{k_t}{p_{t,\text{jet}} \Delta}, \quad (\Delta \ll 1), \quad (2.5)$$

where $C_F = \frac{4}{3}$, $p_{gq}(z) = \frac{1+(1-z)^2}{z}$ ($0 < z < 1$); $\bar{z}$ is an effective momentum fraction and coincides with z in Eq. (2.1) when there is a single emission. For $\bar{z} \ll 1$ the $\bar{z}$ -dependent